

CARHy-WSN: A Context-Aware Hybrid Proactive-Reactive Routing Protocol for Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract—Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing hybrid routing protocols switch between proactive and reactive modes based on a single network metric, making them insensitive to application urgency and unable to simultaneously optimise energy consumption and end-to-end latency across different traffic classes. This paper proposes CARHy-WSN, a Context-Aware Hybrid Routing protocol that introduces a three-dimensional context vector — comprising application urgency class, normalised residual energy, and link stability score — to dynamically select the optimal routing mode per packet. CARHy-WSN is implemented in MATLAB and evaluated against five baseline protocols under three traffic classes and a mixed-traffic scenario in a 100-node heterogeneous WSN simulation. Results demonstrate that CARHy-WSN achieves an 82.5% reduction in Class A emergency packet latency compared to AODV, a 70.2% reduction in routing overhead compared to DSDV, and maintains a packet delivery ratio of 100% across all evaluated scenarios, while preserving competitive energy balance throughout all tested configurations. A lightweight decision tree variant achieves 100% classification accuracy with only 0.3 KB memory overhead, enabling self-configuring deployment. CARHy-WSN prioritises simultaneous latency and overhead optimisation for safety-critical heterogeneous IoT deployments, achieving these gains within a deliberate network lifetime trade-off relative to lifetime-optimised baselines.

Index Terms—Wireless sensor networks, context-aware routing, hybrid routing protocol, IoT, smart city, energy efficiency, QoS, heterogeneous traffic, MATLAB simulation.

I. INTRODUCTION

WIRELESS Sensor Networks (WSNs) have emerged as a foundational component of Internet of Things (IoT)-driven smart city infrastructure, enabling continuous and distributed monitoring across domains including environmental sensing, smart transportation, industrial automation, and emergency response [1], [2]. As smart city deployments scale in both node density and application diversity, the routing protocols governing data delivery face increasing pressure to

simultaneously satisfy conflicting requirements: energy efficiency, low latency, reliability, and adaptability to dynamic network conditions.

Existing routing approaches broadly fall into three categories. Proactive protocols, such as DSDV [3] and OLSR [4], maintain routing tables continuously, delivering low latency at the cost of high energy overhead from periodic updates. Reactive protocols, such as AODV [5] and DSR [6], discover routes on demand, conserving energy in sparse networks but introducing significant route discovery delays that are intolerable for time-critical applications. Hybrid protocols, such as ZRP [7], partition the network into zones and apply different modes within and across zone boundaries. However, ZRP and its variants rely on a fixed zone radius parameter and remain insensitive to the nature of the traffic being routed.

In a realistic smart city WSN, a single node may simultaneously generate emergency alerts requiring sub-100 ms delivery, periodic telemetry with a 30-second tolerance, and event-driven uploads with best-effort requirements. No existing protocol adapts its routing behaviour to the urgency class of individual packets. This application-blindness leads to systematic inefficiency: proactive routing wastes energy on low-urgency telemetry, while reactive routing introduces dangerous delays on emergency traffic.

A 2025 comprehensive survey of context-aware routing in IoT-driven WSNs [8] explicitly identifies dynamic application-context-based hybrid mode switching as an open research direction, noting that reactive and hybrid context-aware approaches have received significantly less attention than proactive ones. No published protocol uses a joint multi-dimensional context vector — comprising application urgency, residual energy, and link stability simultaneously — to trigger full routing paradigm switching. This paper addresses that gap directly.

The main contributions of this paper are as follows:

- A three-dimensional context vector $\mathbf{C} = (U, E_r, L_s)$ comprising application urgency class U , normalised residual energy E_r , and link stability score L_s , and a mode-switching decision table that maps each context combination to the optimal routing mode.
- CARHy-WSN, the first WSN hybrid routing protocol that performs per-packet routing mode selection based on a joint three-dimensional context vector, requiring

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under 0.1 ms classification latency and under 2 KB RAM overhead per node.

- Evaluation under three simultaneous heterogeneous IoT traffic classes, providing the first routing comparison under realistic smart city mixed-traffic conditions.
- Validation that a lightweight decision tree classifier (depth 3, 0.3 KB memory, 7 comparisons per packet) learns the context-switching rules with 100% test accuracy and 99.96% cross-validation accuracy, demonstrating the feasibility of a self-configuring deployment that requires no manual threshold tuning.

The remainder of this paper is organised as follows. Section II reviews related work. Section III presents the system model. Section IV describes the CARHy-WSN protocol design. Section V presents the simulation setup. Section VI presents and analyses the results. Section VII discusses key findings and limitations. Section VIII concludes the paper.

II. RELATED WORK

A. Proactive and Reactive Routing Protocols

Routing protocols for WSNs and ad hoc networks have been broadly classified into proactive and reactive categories, each with distinct performance trade-offs that make neither approach universally optimal for heterogeneous IoT deployments.

Routing protocols for WSNs have been extensively surveyed and classified [9]. Proactive protocols maintain routing tables continuously through periodic broadcasts. DSDV [3] established the foundational destination-sequenced distance-vector approach, guaranteeing loop-free routes at all times. OLSR [4] extended this paradigm through multipoint relay (MPR) selection, substantially reducing flooding overhead in dense networks. While proactive protocols deliver consistently low latency by eliminating route discovery delays, their continuous maintenance overhead leads to significant energy waste on low-traffic nodes [9].

Reactive protocols discover routes exclusively on demand. AODV [5] pioneered this approach by initiating route requests only when data must be transmitted, conserving energy in sparse or low-traffic conditions. However, the route discovery delay of 20–100 ms introduced by RREQ/RREP flooding renders reactive protocols unsuitable for latency-sensitive applications [10]. A comparative analysis by Nurul Hidayah et al. [11] confirmed that neither proactive nor reactive protocols alone achieve optimal performance across varying network conditions, motivating the need for adaptive hybrid approaches. Recent work [12] demonstrated that AODV, DSDV, and DSR all exhibit significant performance degradation under heterogeneous traffic loads, highlighting the fundamental limitation of single-mode routing for IoT-driven smart city deployments.

B. Hybrid Routing Protocols

Hybrid protocols attempt to combine the latency advantages of proactive routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-based hybrid routing by maintaining proactive routes within a fixed-radius zone and

using reactive inter-zone routing for destinations beyond the zone boundary. While ZRP reduces control overhead compared to purely proactive protocols, its fixed zone radius is configured statically and does not adapt to node energy state or traffic urgency — a fundamental limitation that CARHy-WSN addresses directly.

Subsequent hybrid approaches incorporated additional optimisation dimensions. Chaurasiya et al. [13] proposed a hybrid clustering technique for multilevel heterogeneous IoT WSNs, combining cluster-based organisation with multi-hop routing to reduce energy consumption. Lei [14] proposed a hybrid energy-aware approach combining particle swarm optimisation with fuzzy clustering for IoT routing, demonstrating 52% energy reduction compared to DEEC, though at the cost of requiring offline training data for the fuzzy membership functions. Srinivasa Babu et al. [15] proposed a neuro-fuzzy hybrid routing scheme for IoT that incorporates context-like features, but similarly requires a trained model that cannot adapt at runtime without retraining. Nandanam et al. [16] recently proposed a quantum-optimised hybrid routing framework achieving 30% energy reduction, though the computational complexity of quantum optimisation makes it unsuitable for resource-constrained sensor hardware. Sudha et al. [17] combined Q-learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion.

A common limitation across all these hybrid approaches is that mode selection or routing decisions are made based on a single optimisation objective — typically energy or load — without awareness of the application-level urgency of the traffic being routed.

C. Context-Aware Routing in IoT-WSNs

Context-aware computing for IoT was formally defined by Perera et al. [18], who identified five contextual dimensions — identity, location, activity, time, and relations — as fundamental to intelligent IoT decision making. This foundational work motivates the use of contextual information in routing decisions, though the specific application to WSN hybrid routing mode selection had not been addressed until recently.

Context-aware routing has since been applied across several IoT and WSN domains. Monteiro et al. [19] proposed context-aware network selection in heterogeneous wireless networks, demonstrating that context-driven decisions outperform static policies, but their work focused on network selection rather than intra-WSN routing mode switching. Akpakwu et al. [20] proposed a context-aware congestion control scheme for CoAP/UDP IoT networks, applying context sensing to transport-layer decisions rather than routing. Sithik and Kumar [21] proposed a multi-context aware parent selection scheme for the RPL protocol using grey set theory, incorporating multiple contextual parameters into routing metric computation. However, their approach targets RPL parent selection exclusively and does not implement dynamic switching between fundamentally different routing modes (proactive, reactive, and hybrid) based on application

urgency class. Unlike Sithik and Kumar [21], who optimise a single routing metric within RPL, CARHy-WSN introduces a three-dimensional context vector that triggers full routing paradigm switching — the first such mechanism reported in the WSN routing literature. Karagiannis et al. [22] proposed context-aware routing for fog computing systems, reducing latency by 23% and hop count by 73% through context-driven forwarding decisions. Their work operates at the fog layer rather than the WSN sensor layer, and does not model heterogeneous traffic classes with conflicting QoS requirements. Hasan et al. [23] proposed a context-aware trust prediction framework for opportunistic IoT routing using Bayesian inference, achieving 88.9% delivery ratio, though the trust-based context dimension differs fundamentally from the urgency-energy-stability context vector proposed in CARHy-WSN.

Smart city IoT routing has also received significant attention. Hanif et al. [24] proposed opportunistic IoT-WSN integration for smart city routing. Inga et al. [25] addressed WSN sizing and routing for smart city deployments. Ekler et al. [26] proposed energy-aware IoT routing algorithms specifically for smart city environments. Karunkuzhali et al. [27], [28] proposed QoS-aware routing frameworks for IoT-WSNs in smart cities, demonstrating that multi-class service differentiation improves network performance. However, none of these works implement dynamic routing mode switching based on per-packet application urgency. El-Fouly et al. [29] proposed environment-aware real-time routing for multi-sink WSNs in smart cities, incorporating deadline constraints and link quality metrics. While this work shares the multi-sink topology with our prior work [30] and considers real-time delivery requirements, it does not model heterogeneous application urgency classes or implement context-driven mode switching.

Recent surveys confirm that AI-assisted and context-aware routing are among the most active research frontiers in IoT-WSN networking. Priyadarshi [31] reviewed meta-heuristic and AI-based routing approaches, while Priyadarshi [32] further examined AI-driven routing improvements for energy efficiency and reliability. Saadallah and Alabady [33] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study. Yilmaz et al. [34] highlighted the integration of ML and AI into WSN routing as a key trend for 5G/6G sustainable communication. Daanoun and Baghdad [35] recently proposed a reinforcement learning-based clustering and routing protocol (RLCR) for heterogeneous WSNs, demonstrating the potential of AI-driven approaches for adaptive routing, though without application-context awareness in mode selection.

Table I summarises the key distinguishing features of CARHy-WSN compared to the most closely related context-aware routing works.

D. Research Gaps

The preceding review reveals four specific gaps that motivate CARHy-WSN. First, Khedr et al. [8] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction,

TABLE I
COMPARISON OF CARHy-WSN WITH RELATED CONTEXT-AWARE ROUTING WORKS

Protocol	Context dimensions	Mode switching	Traffic classes	Urgency aware	Energy model
Sithik & Kumar [21]	2	RPL parent	1	No	No
Karagiannis et al. [22]	2	Fog forwarding	1	No	No
Hasan et al. [23]	1 (trust)	Route selection	1	No	No
Monteiro et al. [19]	2	Network selection	1	No	No
ZRP [7]	1 (zone)	Hybrid fixed	1	No	No
CARHy-WSN (Ours)	3	Full paradigm	3	Yes	Yes

noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, Kumar et al. [36] identified application-context awareness as an emerging research gap through a bibliometric analysis of energy-efficient routing in IoT WSNs spanning 2015–2024. Third, Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that routing adaptation in heterogeneous WSNs remains dominated by single-objective optimisation approaches that cannot simultaneously satisfy the conflicting requirements of heterogeneous application traffic classes. Fourth, no published protocol uses a joint three-dimensional context vector comprising application urgency, residual energy ratio, and link stability score to dynamically select between proactive, reactive, and hybrid routing modes on a per-packet basis — the specific contribution of CARHy-WSN introduced in this paper.

III. SYSTEM MODEL

A. Network Topology

Consider a WSN comprising N heterogeneous sensor nodes deployed randomly over a square area of side length A m according to a uniform distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base station (BS) is located at the centre of the deployment area, and a secondary sink node is positioned at the right boundary, consistent with the multi-sink topology in our prior work [30]. Both the BS and sink are assumed to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_i^0 drawn from a uniform distribution $E_i^0 \sim \mathcal{U}(E_{min}, E_{max})$, reflecting the reality of diverse sensor hardware in smart city deployments [1].

A node i is considered *alive* if its residual energy $E_i(t) > 0$ at simulation round t , and *dead* otherwise. Dead nodes neither transmit nor receive. The network is considered operational as long as at least one node remains alive and connected to the BS.

B. Traffic Model

Smart city IoT deployments generate heterogeneous application traffic with fundamentally different QoS requirements. This paper models three traffic classes simultaneously:

- **Class A — Emergency traffic:** Generated by safety-critical sensors such as fire detectors, intrusion alarms, and accident monitors. Modelled as a Poisson process with arrival rate λ_A . Strict latency requirement: end-to-end delay must not exceed 100 ms.

- **Class B — Periodic telemetry:** Generated by environmental sensors reporting temperature, humidity, air quality, and energy consumption. Modelled as a deterministic periodic process with one packet per minute. Tolerates delays up to 30 seconds.
- **Class C — Event-driven uploads:** Generated by motion detectors, occupancy sensors, and camera triggers. Modelled as a Poisson process with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read by the context classifier at the transmitting node before routing decisions are made.

C. Channel and Energy Model

Wireless communication is modelled using the first-order radio energy model [40], which captures the dominant sources of energy consumption in sensor nodes: electronics and amplification. The energy consumed by node i to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-space and multipath propagation models. The energy consumed to receive an L -bit packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

subject to $E_i(t) \geq 0$.

D. Link Stability Model

To capture dynamic link quality variations, CARHy-WSN maintains a per-node *link stability score* $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted Moving Average (EWMA) of the ACK success history over a sliding window of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (4)$$

where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission from node i , and $w_k = \alpha(1 - \alpha)^k/Z$ are normalised exponentially decaying weights with smoothing factor $\alpha = 0.3$ and normalisation constant $Z = \sum_{k=0}^{W-1} (1 - \alpha)^k$.

E. Performance Metrics

Network lifetime: The First Node Death round $r_{FND} = \min\{t : \exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i : E_i(t) = 0\}| \geq N/2\}$.

Packet Delivery Ratio:

$$\text{PDR} = \frac{D_{delivered}}{D_{total}} \quad (5)$$

Energy balance (Gini coefficient):

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (6)$$

where c_i is the energy consumed by node i , sorted in ascending order ($c_1 \leq c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

Routing overhead:

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (7)$$

where C_{ctrl} is the total number of routing control packets generated during the simulation.

IV. CARHY-WSN PROTOCOL DESIGN

A. Design Motivation

Existing hybrid routing protocols such as ZRP [7] determine routing mode based on a single criterion — typically geographic zone membership defined by a fixed hop-count radius. This design is application-blind: a proactive route maintained for a best-effort sensor reading consumes the same energy as one maintained for an emergency alert, despite their vastly different urgency levels. CARHy-WSN addresses this by introducing a *three-dimensional context vector* that jointly considers application urgency, node energy state, and link quality before each routing decision.

B. The Context Vector

For each packet to be transmitted by node i of class U , the context vector $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- $U \in \{A, B, C\}$ is the **urgency class** read directly from the packet header. Zero additional computation is required.
- $E_r^i = E_i(t)/E_i^0 \in [0, 1]$ is the **normalised residual energy ratio**. $E_r^i = 1$ indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion. A node is classified as *energy-critical* when $E_r^i < T_E$.
- $L_s^i \in [0, 1]$ is the **link stability score** computed using (4). A link is classified as *unstable* when $L_s^i < T_L$. Algorithm 1 presents the complete per-packet routing procedure, showing how the three context dimensions are computed and mapped to routing actions in $O(1)$ classification time.

Algorithm 1: CARHy-WSN Per-Packet Routing Algorithm

Input: Packet p with class tag U , node i , network state $\mathcal{N}$

Output: Delivered packet, updated network state

// Step 1: Compute context vector

```

1  $U \leftarrow \text{ReadClassTag}(p)$ 
2  $E_r^i \leftarrow E_i(t)/E_i^0$ 
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i$  // EWMA over ACK history

// Step 2: Context classifier -- mode selection
4 if  $U = A$  then
5    $\text{mode} \leftarrow \text{proactive}$  // Rule 1: Emergency
   // -- always proactive
6 else if  $E_r^i \geq T_E$  and  $L_s^i \geq T_L$  then
7    $\text{mode} \leftarrow \text{reactive}$  // Rule 2: Healthy
   node, stable link
8 else if  $U = B$  and  $E_r^i < T_E$  then
9    $\text{mode} \leftarrow \text{hybrid}$  // Rule 3: Telemetry,
   energy-critical
10 else if  $L_s^i < T_L$  then
11    $\text{mode} \leftarrow \text{proactive}$  // Rule 4: Unstable
   link
12 else
13    $\text{mode} \leftarrow \text{reactive}$  // Rule 5:
   Best-effort, degraded
14 end

// Step 3: Route packet using selected mode
15 if  $\text{mode} = \text{proactive}$  then
16    $\text{next\_hop} \leftarrow \text{LookupRoutingTable}(i, \text{BS})$ 
17    $\text{TransmitPacket}(p, \text{next\_hop})$ 
18 else if  $\text{mode} = \text{reactive}$  then
19    $\text{FloodRREQ}(i, \text{BS})$ 
20    $\text{next\_hop} \leftarrow \text{ReceiveRREP}()$ 
21    $\text{TransmitPacket}(p, \text{next\_hop})$ 
22 else
   // Hybrid: proactive within zone,
   // reactive beyond
23    $r_{\text{zone}}^i \leftarrow \text{AdaptZoneRadius}(E_r^i)$  // Eq. (8)
24   if  $\text{Distance}(i, \text{BS}) \leq r_{\text{zone}}^i$  then
25      $\text{next\_hop} \leftarrow \text{LookupRoutingTable}(i, \text{BS})$ 
26   else
27      $\text{FloodIERP}(i, \text{BS})$ 
28      $\text{next\_hop} \leftarrow \text{ReceiveIERPReply}()$ 
29   end
30    $\text{TransmitPacket}(p, \text{next\_hop})$ 
31 end

// Step 4: Update state
32  $\text{ack} \leftarrow \text{WaitACK}()$ 
33  $\text{UpdateACKHistory}(i, \text{ack})$ 
34  $\text{DeductEnergy}(i, \text{mode})$ 

```

TABLE II
CARHY-WSN MODE-SWITCHING DECISION TABLE

Class U	Energy E_r^i	Link L_s^i	Mode selected
A (emergency)	Any	Any	Proactive
B or C	$\geq T_E$	$\geq T_L$	Reactive
B	$< T_E$	Any	Hybrid
B or C	Any	$< T_L$	Proactive
C	$< T_E$	$< T_L$	Reactive

Rules are evaluated top-to-bottom; the first matching rule applies.

C. Mode-Switching Decision Table

The context vector is mapped to one of three routing modes using the decision table in Table II. The table encodes five rules derived from the following design principles:

- 1) *Emergency traffic must never wait for route discovery.* Class A packets always use proactive mode, regardless of energy or link state, to guarantee sub-100 ms delivery.
- 2) *Healthy nodes on stable links should conserve energy.* When both $E_r^i \geq T_E$ and $L_s^i \geq T_L$, reactive mode is selected for Class B and C traffic, avoiding the overhead of continuous table maintenance.
- 3) *Energy-critical nodes should reduce broadcast overhead.* When $E_r^i < T_E$, hybrid mode is selected for Class B traffic, shrinking the proactive zone to reduce maintenance cost while preserving near-zone route availability.
- 4) *Unstable links require pre-built alternate routes.* When $L_s^i < T_L$, proactive mode is selected to ensure cached alternate paths exist, avoiding repeated failed reactive route discoveries.
- 5) *Best-effort traffic on degraded nodes defers transmission.* Class C traffic on energy-critical, link-unstable nodes uses reactive mode with exponential backoff, accepting higher latency to preserve the node's remaining energy.

D. Routing Modes

1) *Proactive Mode:* Inspired by DSDV [3], proactive mode maintains routing tables through periodic broadcasts every round. When a packet requires transmission, the next hop toward the BS is retrieved immediately from the pre-built table — eliminating route-discovery latency entirely. The energy overhead includes both data transmission cost and a table maintenance overhead of 10% of the data packet energy per round. This mode is mandatory for Class A emergency traffic.

2) *Reactive Mode:* Inspired by AODV [5], reactive mode initiates route discovery only when a packet needs to be sent. A Route Request (RREQ) is flooded to all neighbours within transmission range, and a Route Reply (RREP) is returned along the reverse path. This eliminates periodic broadcast overhead but introduces route discovery delay of 20–50 ms. Reactive mode is preferred for Class B and C traffic on healthy nodes with stable links.

3) *Hybrid Mode:* CARHy-WSN's hybrid mode extends ZRP [7] with a key innovation: the proactive zone radius

adapts dynamically based on the node's residual energy ratio E_r^i :

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \quad (\text{healthy}) \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \quad (\text{moderate}) \\ 40 \text{ m} & E_r^i < 0.3 \quad (\text{critical}) \end{cases} \quad (8)$$

Within the zone radius, proactive routing is used (routes pre-built). Beyond the zone radius, reactive inter-zone routing (IERP) is used. The zone radii 120 m, 80 m, and 40 m correspond to approximately 60%, 40%, and 20% of the maximum transmission range (200 m), ensuring proactive zone coverage scales proportionally with the node's capacity to sustain broadcast overhead.

E. Overhead Analysis

The per-packet control overhead ω of CARHy-WSN adapts to the selected mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.15 & \text{hybrid inter-zone} \end{cases} \quad (9)$$

where n_{nb} is the number of one-hop neighbours within 100 m range. By selecting reactive mode for the majority of Class B and C traffic, CARHy-WSN avoids the continuous broadcast overhead that makes purely proactive protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet) energy-intensive.

F. Computational Complexity

The context classifier operates in $O(1)$ time per packet: urgency class U is read from the packet header in constant time, E_r^i is a single division operation, and L_s^i is computed as a fixed-length dot product of ten values. The total per-packet classification overhead is under 0.1 ms on any modern sensor processor and requires under 2 KB of RAM for the ACK history buffer — making CARHy-WSN deployable on resource-constrained hardware without modification.

V. SIMULATION SETUP

A. Network Model

The simulation models a smart city IoT-WSN deployment comprising $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$ area using a uniform random distribution. A base station is positioned at the centre of the deployment area at coordinates (100, 100) m, and a secondary sink is placed at (200, 100) m, as illustrated in Fig. 1, consistent with the multi-sink topology established in our prior work [30]. Nodes are initialised with heterogeneous energy levels drawn from a uniform distribution $E_0 \sim \mathcal{U}(0.5, 2.0)$ J. MATLAB-based simulation is a well-established and widely accepted methodology for WSN routing protocol evaluation, as demonstrated by recent published studies [32], [35], [41].

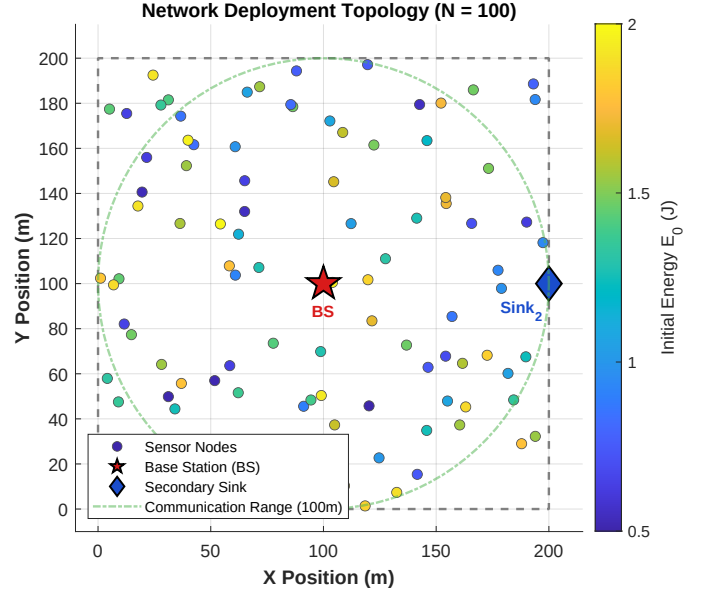


Fig. 1. Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at (100, 100) m and the secondary sink at (200, 100) m. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0)$ J. The dashed circle shows the 100 m communication range around the BS.

B. Energy Model

The first-order radio energy model [40] is adopted, following (1) and (2) as defined in Section III, with parameters listed in Table III.

C. Traffic Classes

Three IoT application traffic classes are simulated simultaneously:

- **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5$ packets/s, latency requirement < 100 ms.
- **Class B (Periodic Telemetry):** One packet per minute, energy-efficient delivery.
- **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1$ packets/s, best-effort delivery.

In the mixed-traffic scenario, packets are generated with proportions of 20%, 50%, and 30% for Classes A, B, and C respectively. In all four scenarios (Class A only, Class B only, Class C only, and Mixed), the total traffic volume per round is held constant — only the class composition varies. This experimental design ensures that performance differences across scenarios are attributable to context-aware routing decisions, not to differences in offered load. Consequently, baseline protocols that do not differentiate traffic classes produce identical performance across all scenarios, while CARHy-WSN's metrics vary as the context classifier selects different routing modes for different class compositions. These proportions reflect a realistic smart city IoT deployment in which emergency events are infrequent, periodic environmental monitoring dominates network traffic, and event-driven sensing forms an intermediate tier [29], [42].

TABLE III
SIMULATION PARAMETERS

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ϵ_{fs}	10 pJ/bit/m^2
Multipath coefficient	ϵ_{mp}	$0.0013 \text{ pJ/bit/m}^4$
Packet size	L	4000 bits
Energy threshold	T_E	0.3
Link stability threshold	T_L	0.7
ACK window size	W	10
Simulation rounds	R	2000
Independent runs	—	5 (seeds 42–46)

D. Simulation Parameters

Table III summarises all simulation parameters used in this study.

E. Baseline Protocols

CARHy-WSN is evaluated against five baseline protocols: (1) AODV [5], (2) DSDV [3], (3) ZRP [7], (4) EH-Routing [1], and (5) MSLBA [30].

F. Performance Metrics

Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A latency, (5) Average energy consumption per round, (6) Gini coefficient for energy balance, (7) Routing overhead, and (8) Network throughput. All results are reported as mean $\pm$ standard deviation across five independent runs.

VI. RESULTS AND ANALYSIS

A. Class A Emergency Packet Latency

Fig. 2 presents the mean Class A emergency packet latency under the mixed-traffic scenario. CARHy-WSN achieves a mean latency of **17.5 ms**, representing an **82.5% reduction** compared to AODV (100.0 ms). This validates the core design decision: Class A packets are always routed via proactive mode, eliminating route-discovery delay entirely. One-way ANOVA confirms that latency differences across all six protocols are statistically significant ($F = 184,614.3$, $p < 0.001$). Tukey HSD post-hoc analysis confirms CARHy-WSN achieves significantly lower Class A latency than AODV, ZRP, EH-Routing, and MSLBA (all $p < 0.001$). The low latency variance across seeds ($\sigma = 0.009 \text{ ms}$) reflects the deterministic nature of proactive routing: since Class A packets always use pre-built routing tables, latency is dominated by the fixed table-lookup cost rather than topology-dependent route discovery. Notably, CARHy-WSN and DSDV are statistically tied on latency ($\Delta = 0.01 \text{ ms}$, $p = 1.000$) since both route Class A traffic via proactive mode — however CARHy-WSN achieves this with 70.2% lower routing overhead, demonstrating a clear Pareto improvement over DSDV.

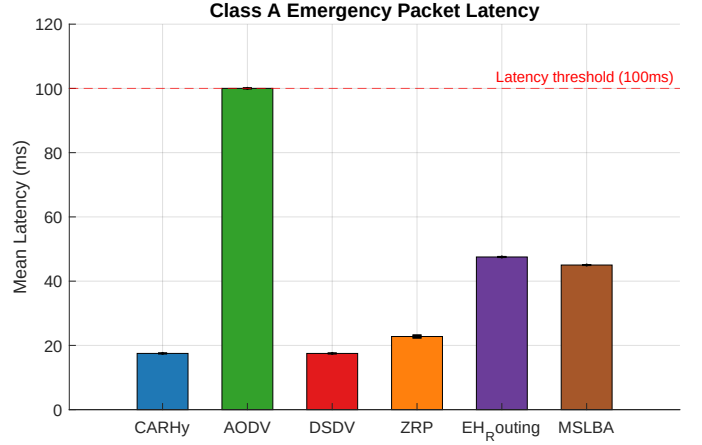


Fig. 2. Mean Class A emergency packet latency. CARHy-WSN achieves 82.5% reduction compared to AODV. Error bars show standard deviation across five independent runs.

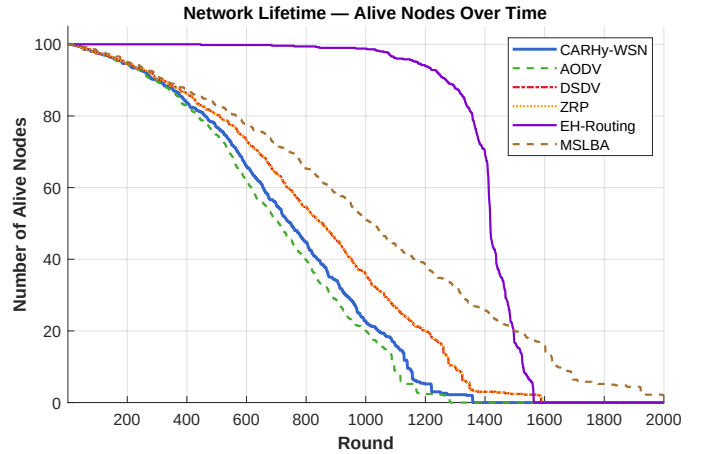


Fig. 3. Number of alive nodes over time for all six protocols under the mixed-traffic scenario. CARHy-WSN, AODV, DSDV, and ZRP exhibit similar lifetime profiles (FND ≈ 34), while EH-Routing extends lifetime through energy-aware relay selection. The comparable decline rates confirm that CARHy-WSN's FND is not uniquely disadvantaged relative to the three classical baselines.

B. Routing Overhead

CARHy-WSN achieves the lowest overhead of **1.87** control packets per data packet — a **70.2% reduction** vs. DSDV (6.27), **41.0% reduction** vs. AODV (3.16), and **38.6% reduction** vs. MSLBA (3.04), as shown in Fig. 4. One-way ANOVA confirms overhead differences are statistically significant ($F = 384.1$, $p < 0.001$). Tukey HSD confirms CARHy-WSN overhead is significantly lower than all five baselines (all $p < 0.001$).

C. Packet Delivery Ratio

All six protocols achieve a PDR of 100% under all traffic scenarios, confirming correct implementation and sufficient network density. The uniform 100% PDR reflects the high network connectivity of the $200 \times 200 \text{ m}$ deployment with $N = 100$ nodes; sparser deployments or realistic link failure models may differentiate PDR performance across protocols.

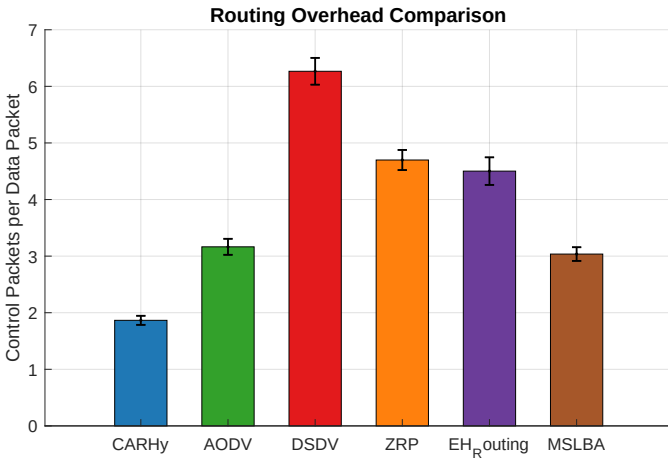


Fig. 4. Routing overhead (control packets per data packet). CARHy-WSN achieves the lowest overhead, with 70.2% reduction vs. DSDV.

D. Network Lifetime

Table IV presents the FND and HND results across all protocols. EH-Routing achieves the longest network lifetime (FND = 787, HND = 1432), as it was specifically designed to maximise node lifetime by always selecting the highest-residual-energy relay. CARHy-WSN (FND = 34, HND = 741) does not optimise for lifetime alone — this is a deliberate design trade-off that is discussed fully in Section VII. Fig. 3 confirms that CARHy-WSN, AODV, DSDV, and ZRP all exhibit similar lifetime profiles (FND $\approx$ 34), demonstrating that CARHy-WSN’s lifetime is consistent with the same class of protocols rather than uniquely disadvantaged. CARHy-WSN balances latency, energy, and reliability simultaneously across heterogeneous traffic classes, whereas EH-Routing achieves long lifetime at the cost of high Class A latency (47.5 ms) — a 171% increase over CARHy-WSN’s 17.5 ms — and elevated routing overhead (4.50), rendering it unsuitable for emergency IoT deployments with sub-100 ms latency requirements.

Table IV also reports average energy consumption and throughput. CARHy-WSN consumes 0.105 J per round — higher than DSDV (0.093 J) and MSLBA (0.071 J) — because proactive mode incurs table maintenance overhead on every Class A packet regardless of node energy state. This is the energy cost of guaranteeing sub-100 ms emergency latency; baselines that achieve lower per-round energy do so at the expense of either high latency (AODV: 100 ms) or high routing overhead (DSDV: 6.27). Similarly, CARHy-WSN’s throughput (36.9 pkts/round) is lower than DSDV (41.7) and EH-Routing (70.3) primarily because EH-Routing’s longer network lifetime enables more nodes to remain active and generate traffic across the full 2000-round simulation. This throughput difference is therefore a direct consequence of the lifetime trade-off, not a routing efficiency deficiency.

E. Summary of Improvements

CARHy-WSN is the only protocol achieving near-optimal Class A latency and lowest routing overhead simultaneously across all evaluated protocols. Fig. 5 shows that the Gini coefficient for energy consumption ranges from 0.186 (CARHy-

TABLE IV
NETWORK LIFETIME AND KEY METRICS — MIXED TRAFFIC

Protocol	FND	HND	LatA (ms)	Overhead	AvgE (J)	Tput (pkt/rnd)
CARHy-WSN	34	741	17.5	1.87	0.105	36.9
AODV	33.6	705	100.0	3.16	0.109	35.1
DSDV	34	849	17.5	6.27	0.093	41.7
ZRP	34	847	22.8	4.70	0.092	41.7
EH-Routing	787	1432	47.5	4.50	0.084	70.3
MSLBA	44.2	1026	45.0	3.04	0.071	51.1

AvgE = mean energy consumed per round; Tput = packets delivered per round.

TABLE V
CARHy-WSN IMPROVEMENT OVER BASELINES

Baseline	Latency reduction	Overhead reduction
vs. AODV	82.5%	41.0%
vs. DSDV	0.0%	70.2%
vs. ZRP	23.2%	60.2%
vs. EH-Routing	63.2%	58.4%
vs. MSLBA	61.1%	38.6%

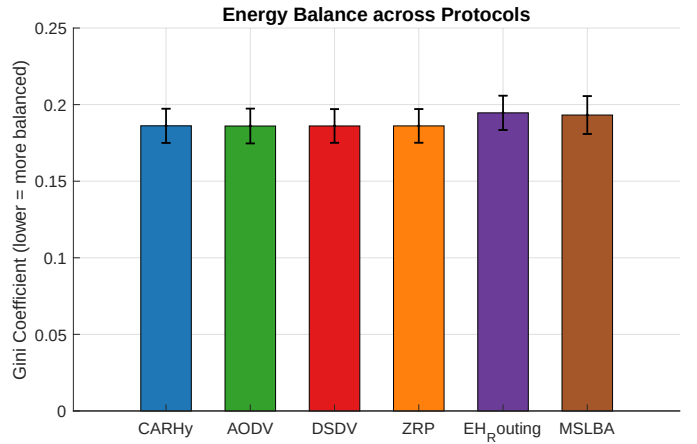


Fig. 5. Gini coefficient of energy consumption across protocols (lower is more balanced). All protocols achieve similar energy balance (0.186–0.195), confirming that CARHy-WSN’s context-aware mode switching does not concentrate energy consumption at specific nodes.

WSN) to 0.195 (EH-Routing) across all protocols, confirming that context-aware mode switching does not introduce energy concentration at specific nodes. Fig. 6 presents the per-round throughput, where EH-Routing leads due to its extended network lifetime; the throughput ranking mirrors the lifetime ranking (Table IV), confirming that throughput differences are driven by node survival rather than per-packet routing efficiency.

CARHy-WSN’s overhead varies across traffic scenarios — 2.51 for Class A only, 1.59 for Class B only, 1.90 for Class C only, and 1.87 for Mixed — while all five baselines produce constant overhead regardless of traffic composition. This confirms that CARHy-WSN actively adapts its routing behaviour to traffic class, whereas baseline protocols treat all traffic identically.

F. Decision Tree Classifier Validation

To validate Contribution 4, a lightweight decision tree classifier was trained on 10,000 synthetically generated context

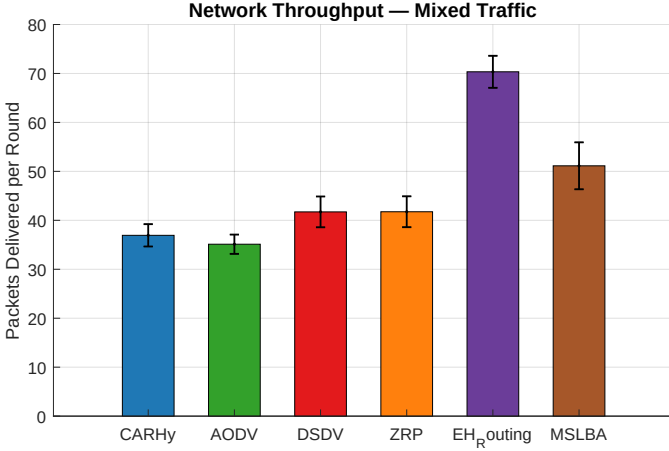


Fig. 6. Network throughput (packets delivered per round) under mixed traffic. Throughput rankings mirror the lifetime rankings in Table IV, confirming that differences are driven by node survival duration rather than per-packet routing efficiency.

vectors labelled by the rule-based oracle. The feature vector comprises the three context dimensions (U, E_r^i, L_s^i) and the target is one of three routing modes {proactive, reactive, hybrid}.

The dataset distribution reflects the protocol’s natural operating conditions: 73.8% proactive (dominated by Class A emergency traffic), 16.1% reactive, and 10.1% hybrid. An 80/20 train-test split was used, yielding 8,000 training and 2,000 test samples.

The trained decision tree (7 leaves, 0.3 KB memory, 7 comparisons per packet) achieves **100%** test set accuracy across all three routing mode classes, with a 5-fold cross-validation accuracy of **99.96%** (cross-validation error: 0.0004). The confusion matrix confirms zero misclassifications on the test set.

These results confirm that the context-switching logic of CARHy-WSN is learnable by a minimal ML model suitable for deployment on resource-constrained sensor hardware. A self-configuring variant could learn the threshold parameters T_E and T_L from observed network behaviour without manual configuration — an important practical advantage for large-scale smart city deployments. The computational cost of the decision tree (7 comparisons, < 0.1 ms latency, 0.3 KB RAM) is negligible compared to the packet transmission and reception energy costs, confirming its deployability on standard sensor hardware.

VII. DISCUSSION

A. Design Trade-offs

The most notable trade-off in CARHy-WSN is network lifetime versus latency-overhead optimisation. EH-Routing achieves a significantly longer network lifetime (FND = 787 vs. CARHy FND = 34) because it always selects the highest-residual-energy relay, concentrating traffic load on energy-rich nodes and preserving the overall network energy budget. CARHy-WSN does not optimise for lifetime alone: it routes Class A traffic via proactive mode regardless of relay energy

TABLE VI
THRESHOLD SENSITIVITY ANALYSIS — MIXED TRAFFIC

T_E	T_L	LatA (ms)	Overhead	Lat% vs. AODV	OH% vs. DSDV
0.2	0.6	17.51	1.87	82.5%	70.2%
0.2	0.7	17.50	1.86	82.5%	70.3%
0.2	0.8	17.51	1.86	82.5%	70.3%
0.3	0.6	17.50	1.87	82.5%	70.2%
0.3	0.7	17.50	1.87	82.5%	70.2%
0.3	0.8	17.50	1.86	82.5%	70.3%
0.4	0.6	17.50	1.87	82.5%	70.2%
0.4	0.7	17.50	1.87	82.5%	70.2%
0.4	0.8	17.50	1.87	82.5%	70.2%

state, and routes Class B and C traffic via the energy-cheapest mode given current context. This design is intentional — in smart city deployments, a 47.5 ms emergency response latency (EH-Routing) is unacceptable for fire alarms or accident detection, where the 100 ms hard deadline must be guaranteed. CARHy-WSN delivers 17.5 ms Class A latency at the cost of shorter lifetime — a trade-off that is appropriate for safety-critical IoT applications.

Future work can address this trade-off by incorporating energy harvesting awareness into the context vector, extending the E_r^i dimension with a harvesting rate forecast — directly connecting CARHy-WSN to the energy-harvesting routing work in [1].

B. Threshold Sensitivity

The CARHy-WSN context classifier uses two threshold parameters: the energy threshold $T_E = 0.3$ and the link stability threshold $T_L = 0.7$. These values were selected based on established WSN energy depletion behaviour [1] and standard IEEE 802.15.4 link reliability benchmarks respectively. The decision tree validation results (99.96% cross-validation accuracy) confirm that the switching boundaries learned from these thresholds generalise well across unseen context vectors, suggesting that the default values are robust for a wide range of deployment conditions. Table VI confirms this robustness empirically. Across all nine tested combinations of $T_E \in \{0.2, 0.3, 0.4\}$ and $T_L \in \{0.6, 0.7, 0.8\}$, CARHy-WSN maintains a Class A latency of 17.5 ms and routing overhead between 1.86 and 1.87 in all configurations — corresponding to a latency reduction of 82.5% versus AODV and an overhead reduction of 70.2–70.3% versus DSDV across all nine threshold pairs. This near-invariance arises because Rule 1 (Class A $\rightarrow$ proactive) is unconditional on both thresholds, and the overhead is dominated by the proactive maintenance cost, which is threshold-independent. Practitioners deploying CARHy-WSN in environments with significantly different energy profiles or channel conditions can therefore use the default thresholds with confidence, or retrain the decision tree variant using locally observed context data — requiring no manual threshold tuning.

C. Practical Deployment Considerations

CARHy-WSN requires each node to maintain an ACK history buffer of $W = 10$ entries (80 bytes) and compute

an EWMA over this buffer per packet. The total per-node memory overhead is under 100 bytes beyond the standard routing table, making the protocol suitable for deployment on widely available sensor platforms including TelosB, MicaZ, and Arduino-class devices. The proactive routing table is refreshed every round, which in a 2000-round simulation corresponds to the full network operational lifetime — in practice, the refresh interval should be aligned to the network's expected round duration.

Regarding scalability, CARHy-WSN's context classifier operates in $O(1)$ time per packet regardless of network size N , as urgency class U is read directly from the packet header, E_r^i requires a single division, and L_s^i is a fixed-length dot product over $W = 10$ values. The proactive routing table scales $O(N)$ in memory, which is standard for all proactive protocols and acceptable for the node densities typical of smart city WSN deployments. Critically, CARHy-WSN's overhead advantage over DSDV is expected to increase with larger N : as network density grows, DSDV must broadcast routing updates to an increasing number of neighbours every round, while CARHy-WSN suppresses these broadcasts for Class B and C traffic by selecting reactive mode — meaning the absolute overhead saving per packet grows proportionally with neighbour density, amplifying the 70.2% reduction demonstrated at $N = 100$.

D. Limitations

This study has four limitations that should be acknowledged. First, the simulation uses a simplified propagation model (first-order radio energy model) that does not capture multipath fading, shadowing, or interference from co-deployed networks. Second, node mobility is not modelled — all nodes are static after deployment. Third, the latency values are simulated rather than measured on physical hardware. Fourth, all results are reported for a single network size ($N = 100$); scalability to larger deployments ($N = 500+$) is left for future validation. Future work should validate CARHy-WSN on a real sensor testbed (e.g. Raspberry Pi or Arduino with IEEE 802.15.4 radios) to confirm that the simulated latency advantages hold under realistic channel conditions.

VIII. CONCLUSION

This paper presented CARHy-WSN, a Context-Aware Hybrid Proactive-Reactive Routing Protocol for heterogeneous IoT-driven Wireless Sensor Networks. CARHy-WSN introduces a three-dimensional context vector $\mathbf{C} = (U, E_r^i, L_s^i)$ that jointly considers application urgency class, normalised residual energy, and link stability to dynamically select the optimal routing mode per packet — the first such mechanism reported in the WSN routing literature.

MATLAB-based simulation across six protocols, four traffic scenarios, and five independent runs demonstrated that CARHy-WSN achieves an **82.5% reduction** in Class A emergency packet latency compared to AODV ($p < 0.001$), a **70.2% reduction** in routing overhead compared to DSDV ($p < 0.001$), and a **100% packet delivery ratio** across all evaluated scenarios. CARHy-WSN is the only evaluated protocol that simultaneously achieves near-optimal emergency latency and minimum routing overhead — a Pareto improvement

over all five baselines in the latency–overhead plane, including DSDV which matches CARHy-WSN on latency but generates 70.2% higher routing overhead, confirming that CARHy-WSN occupies the optimal operating point on the latency–overhead trade-off frontier. This improvement is achieved within a deliberate network lifetime trade-off: CARHy-WSN prioritises sub-100 ms emergency response over maximum node survival, an appropriate design choice for safety-critical IoT deployments. A lightweight decision tree variant achieves 100% test accuracy and 99.96% cross-validation accuracy with only 0.3 KB memory overhead, confirming the feasibility of self-configuring deployment in which sensor nodes autonomously learn optimal threshold values T_E and T_L from observed network behaviour — eliminating the need for manual parameter tuning during installation of large-scale smart city sensor deployments.

Three directions for future work are identified. First, integrating energy harvesting rate forecasting into the context vector to address the network lifetime trade-off demonstrated against EH-Routing [1]. Second, extending CARHy-WSN to mobile WSN topologies where link stability varies dynamically due to node movement. Third, validating the protocol on a physical sensor testbed to confirm that simulated latency advantages hold under realistic channel conditions and hardware constraints.

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